

Analysis and design of simple low-loss wide-angle waveguide bends

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Abstract

In this paper a simple low-loss planar waveguide wide-angle bend is proposed and analyzed. The waveguide bend of this work consists of a curved tapered angle such that its tapering can be adjusted for a minimum loss (maximum transmission coefficient) and small dimension of bend. The analysis is done using wide-angle BPM (beam propagation method) type Pade (4,4). The simulated results of different bends show that at the same transmission coefficient the area required for the proposed bends are smaller than those of the abrupt bends.

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1. Introduction

Planar waveguide bends are basic elements in integrated optical systems. The bends in dielectric planar waveguides convert the guided modes into the radiation modes, where this conversion introduces a loss in the power of the propagating wave. In principle, due to a considerable geometrical path difference between the inner and outer edges of a wide-angle bend (AA' and BB' in Fig. 1(a)), the beam passing through these paths encounter a local phase difference after they passed the bend. Thus, in planar waveguide the wide-angle bends cause the local phase front of the propagating mode to rotate and this creates a phase-front mismatch. Despite the lack of a specific definition for wide-angle bends, the higher loss in bends with angles greater than 4° , justifies that angles greater than 4° to be referred to as wide angles [1]. In order to analyze the efficiency of the bends quantitatively, the normalized transmission coefficient denoted by η_s is defined. The parameter η_s is given as the ratio of the power of guiding mode propagating out from the bend (P_{out}) to the power of guiding mode entering into the bend (P_{in}). Now, it can be written: $\eta_s = P_{out}/P_{in}$ and the power loss is

$P_L = P_{in} - P_{out}$. However, in addition to the angle, several other parameters such as wavelength, waveguide width and the refractive indices affect the normalized transmission coefficient of the bends. In order to reduce the radiation loss, the angle of the waveguide bend should be small, but smaller angles make the device longer for certain displacement of the waveguides. Therefore, having low-loss wide-angle bends is vital for the design of compact integrated optics. Recently, designing a desirable low-loss wide-angle waveguide bend has become a challenging issue [1–4].

A recent survey shows two main categories of waveguide bends: (i) circular curved bends and (ii) abrupt bends. Unlike abrupt bend, in a circular bend, the bending angle of the waveguide increases very gradually like a segment of a circle, and the width of the waveguide can be adjusted to make either a tapered or a constant width bending [3–5], where it is not the same in an abrupt bending. Four well known types of abrupt bends each with its own properties and behavior are: the chamfered type [6–8], the coupled type [9], the outrigger type [9] and the micro-prism type [10,11].

There are two types of abrupt bends: Fig. 1(a) shows a conventional bend, and Fig. 1(b) shows a tapered one, where the latter is proposed by Han et al. [12], and we call it “linear tapered bending”. The linear tapered bending is

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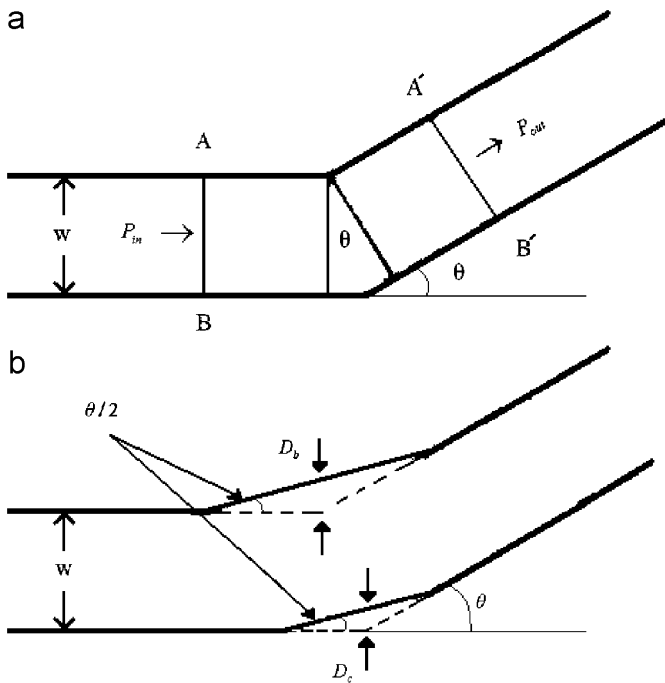


Fig. 1. Illustration of the planar waveguides: (a) conventional bending, (b) the linear tapered bending proposed by Han et al. [12].

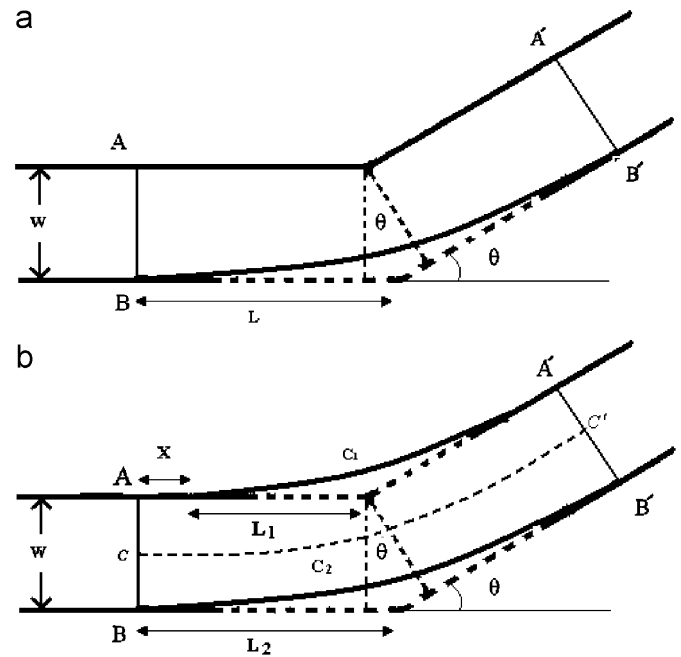


Fig. 2. Illustration of the planar waveguides: (a) single curved bending and (b) double curved bending.

bulged with the core material at the angle of $\theta/2$ at the inner bend edge and the chamfered core region at the angle of $\theta/2$ at the outer bend edge. This is done to reduce the effect of phase-front mismatch within the bend [12]. The linear tapered bending structure shown in Fig. 1(b) provides an increase in the normalized transmission coefficient if compared with that of a conventional structure (Fig. 1(a)). In the “linear tapered bending” the parameter D_b and D_c are called bulged displacement and chamfered displacement, respectively, where these are the design parameters and can be adjusted for the greatest normalized transmission coefficient. Lower D_b and D_c make this structure closer to the abrupt bending structure and lowers the normalized transmission coefficient [12]. At the optimized values of D_b and D_c , the maximum result of the normalized transmission power of this structure are given by: 0.994, 0.99 and 0.923 at the bend angles of 3° , 5° and 7° , respectively [12]. However, this type of bending makes the device relatively large, e.g. at the angles of 3° , 5° and 7° , the bend lengths are 343, 410 and $389\mu\text{m}$, respectively. Note that the bend length is, in fact, the length of a section of the waveguide where the wave front is not perpendicular to the waveguide axis.

In this paper we have proposed a simple structure that significantly increases the normalized transmission coefficient with respect to that of a conventional bending while the length of the proposed structure is also reduced. Two kinds of the proposed bending structures are illustrated in Fig. 2, where we call the structure in Fig. 2(a) the single curved bend and the structure in Fig. 2(b) the double curved bend.

2. Analysis and design

There are descriptions on the geometry and behavior of the proposed structures shown in Fig. 2. In the single curved structure depicted in Fig. 2(a), the edges of the waveguide at the bending corner are unparallel, i.e. this structure deploys a tapered form of the waveguide width at the bending corner. In this structure, the curve BB' is a segment of a circle and the amount of tapering depends on the radius of the circle, R . In the single curved bend (Fig. 2(a)), L is the design parameter which can be adjusted for the maximum normalized transmission coefficient, where it is related to the radius R and angle θ as: $L = R \tan(\theta/2)$. It is noticeable that the increase in the radius of the curve BB' in this structure increases the length of the device and the designer can compromise between the length and the maximum normalized transmission coefficient of the device. Meanwhile, comparison of the structures in Figs. 2(a) and 1(b) indicates that the structure in Fig. 2(a) due to a more smooth turning can apparently reduce the optical mode transfer. In fact, the mode transfer happens at the corner of the structure in Fig. 1(b) due to a noticeable increase in the width of the waveguide at the corner.

In the double curved structure depicted in Fig. 2(b), the curves AA' and BB' are segments of two circles of radii R_1 and R_2 , respectively. In the case of coincident centers of the circles, the edges of the waveguide at the bending corner are parallel. In this case, $R_2 - R_1 = W$ and $L_2 - L_1 = (R_2 - R_1) \tan(\theta/2) = W \tan(\theta/2)$, and the angle of bending increases gradually and the width of the waveguide remains

constant. On the other hand, in the double curved bending if the centers of the circles are not coincident, the value of $(L_2 - L_1)$ is not equal to $W \tan(\theta/2)$. Therefore, the curves AA' and BB' run unparallel and the waveguide at the bend becomes tapered. Setting L_1 and L_2 for a desired tapering of the waveguide at the bending corner can be performed to minimize the radiation loss and maximize the normalized transmission coefficient.

On the other hand, the length of the device depends directly on the values of L_1 and L_2 . Therefore, L_1 and L_2 can be used as the design parameters to be adjusted for a useful compromise between a higher normalized transmission coefficient and a lower device length.

Now, in order to investigate the bending behavior analytically, we derive an equation to calculate the phase difference between the optical waves reaching the points A' and B' with respect to each other. The phase difference between the waves at A' and B' is due to the different geometrical path lengths between AA' and BB' , as shown in Fig. 2.

From the phase point of view, the wave front before bend is AB and after bend is $A'B'$ as shown in Figs. 1 and 2. The geometrical path length difference between the inner path AA' and the outer path BB' is denoted by Δd . The value of Δd and the corresponding phase difference (sometimes called phase mismatch) $\Delta\Phi$ for a conventional abrupt structure shown in Fig. 1(a) are as follows [1]:

$$\Delta d = 2W \tan\left(\frac{\theta}{2}\right), \quad (1a)$$

$$\Delta\Phi = 2\beta W \tan\left(\frac{\theta}{2}\right), \quad (1b)$$

where θ is the bend angle, W is the width of the waveguide and β is the propagation constant in the waveguide. In conventional abrupt bends, due to the phase mismatch, a large amount of power in the guided mode transfers to the radiation mode and scatters. As most of the optical power is confined within the waveguide, it is reasonable to consider the phase mismatch as indication of power radiation within the bend [1]. In the double curved bending, the geometrical path length difference between the inner path AA' and the outer path BB' , Δd is determined as

$$\Delta d = C_2 - C_1 - 2X, \quad (2)$$

where C_1 and C_2 are the inner and the outer arches bending lengths, and X are shown in Fig. 2(b). We can derive X , C_1 and C_2 as follows:

$$X = L_2 - L_1 - W \tan\left(\frac{\theta}{2}\right), \quad (3a)$$

$$C_1 = L_1 \theta \cot\left(\frac{\theta}{2}\right), \quad (3b)$$

$$C_2 = L_2 \theta \cot\left(\frac{\theta}{2}\right). \quad (3c)$$

Replacing (3) into (2), we can obtain the geometrical path length difference Δd and the phase difference $\Delta\Phi$ between the inner and the outer paths of a double curved bending as

$$\Delta d = (L_2 - L_1) \left[\theta \cot\left(\frac{\theta}{2}\right) - 2 \right] + 2W \tan\left(\frac{\theta}{2}\right), \quad (4a)$$

$$\Delta\Phi = \beta(L_2 - L_1) \left[\theta \cot\left(\frac{\theta}{2}\right) - 2 \right] + 2\beta W \tan\left(\frac{\theta}{2}\right). \quad (4b)$$

In Eq. (4b), we observe that the phase difference between the inner and the outer paths in Fig. 2(b) is lower than that of the conventional structure in Fig. 1(a) which is $2\beta W \tan(\theta/2)$. This is because $\theta \cot(\theta/2) < 2$ for typical value of θ . Therefore, we can adjust L_1 and L_2 so that $\Delta\Phi$ becomes optimum and consequently can maximize the normalized transmission coefficient, where in a conventional structure, not any optimization can be performed.

On the other hand, consider the central points of the field within the bending, the path CC' which turns gradually as shown in Fig. 2(b). The geometrical path passing through the center of the waveguide, CC' , can be calculated as

$$CC' = \left[(L_1 - L_2) + (L_2 + L_1) \theta \cot\left(\frac{\theta}{2}\right) + W \tan\left(\frac{\theta}{2}\right) \right], \quad (5)$$

where the value of $\theta \cot(\theta/2)$ is approximately equal to 2 for angles $\theta \leq 10^\circ$. Thus, Eq. (5) reduces to

$$CC' = 3L_1 + L_2 + W \tan\left(\frac{\theta}{2}\right). \quad (6)$$

In Eq. (6) it is seen that for reducing the effect of θ on the geometrical length the values of L_1 and L_2 must be chosen large enough, so that the term $W \tan(\theta/2)$ can be eliminated. In this situation the relative phase change along the central path CC' is given as

$$\Phi = \beta(3L_1 + L_2). \quad (7)$$

Thus, the relative phase change will be independent of θ , when L_1 and L_2 are large. Eventually, the designer can adjust L_1 and L_2 for optimum $\Delta\Phi$ in Eq. (4b).

3. Numerical simulation and analysis

A numerical program is coded to simulate and analyze the device properly and calculate the bending loss. The simulator is based on wide-angle beam propagation method (BPM), and is of the type of Pade (4,4). Pade (4,4) is a well known approximant operator for the type and order of the approximation applied to the calculation of differential equation of the propagating wave in non-paraxial optical devices [14–16]. Using this simulator, it is also possible to analyze the field distribution and phase variation of the wave propagating along the waveguide and through the bending section. We know that at a larger bending θ , the power loss of the optical signal increases and consequently a lower power passes through the bend. As mentioned in Section 1, a measure to characterize the

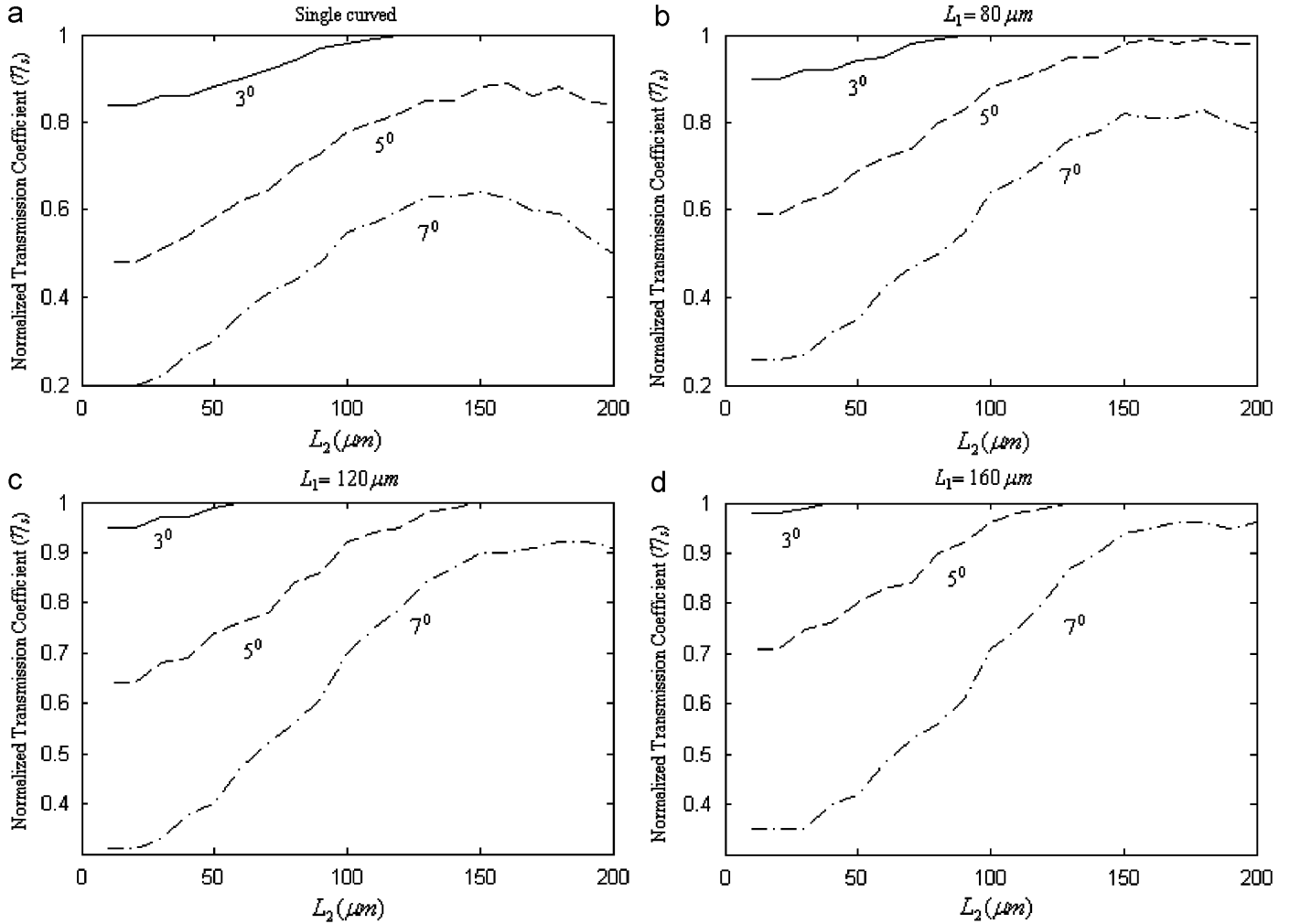


Fig. 3. The illustration of η_s versus L_2 . (a) Simulated η_s versus L_2 (single curved), (b) η_s versus L_2 ($L_1 = 80 \mu\text{m}$), (c) η_s versus L_2 ($L_1 = 120 \mu\text{m}$) and (d) η_s versus L_2 ($L_1 = 160 \mu\text{m}$).

bending quality is called the normalized transmission coefficient (η_s) [13] which is the ratio of the optical power in the guided modes at the end of the bend, $A'B'$ to its value at the beginning of the bend, AB .

The double curved bending shown in Fig. 2(b) is simulated numerically at different bending angles. The dimension and parameters of the device are: $W = 5.5 \mu\text{m}$, $\lambda = 1.55 \mu\text{m}$, $n_s = 1.444$ (substrate) and $n_g = 1.45151$ (guide). The normalized transmission coefficients η_s , of the structure versus L_2 are calculated and the results are illustrated in Fig. 3. In the case of $L_1 = 0$, increase of L_2 increases the normalized transmission coefficient as it is shown in Fig. 3(a). It is interesting to note that in all of the graphs shown in Fig. 3, increase in L_2 lowers the difference between η_s of different θ 's, i.e. in each part of Fig. 3 the curves get closer as L_2 increases which confirms Eq. (7). However, as it is shown in Fig. 2(b), when L_2 increases the waveguide becomes tapered and the bending entrance becomes narrow. Thus, according to Eq. (4b) the phase difference $|\Delta\Phi|$ gets larger than its optimum value and the normalized transmission coefficient, η_s starts to reduce.

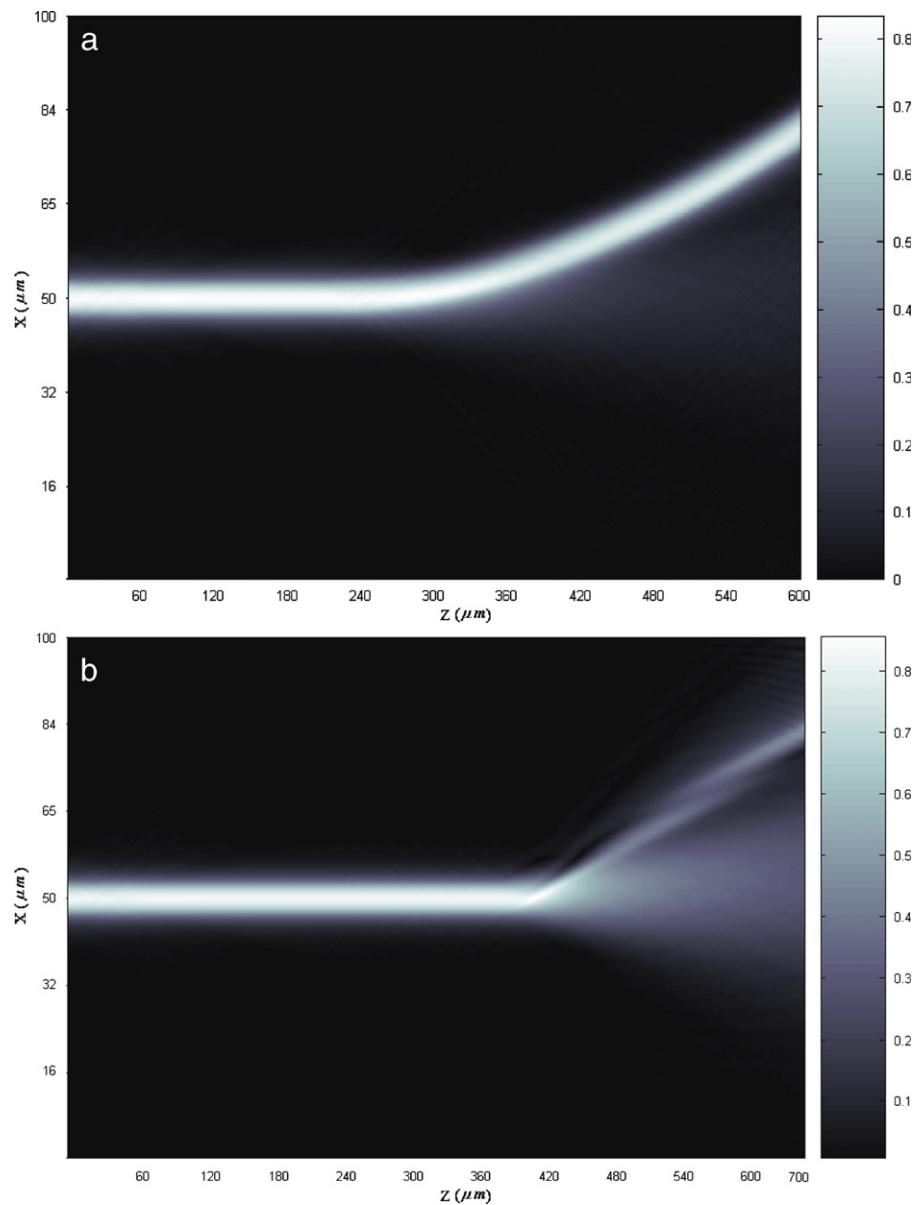
Results of both cases of single curved bending (shown in Fig. 3(a)) and double curved bending ($L_1 \neq 0$) (shown in Fig. 3(b)–(d)) represent a trade-off between the effect of L_2 on η_s through the phase difference ($\Delta\Phi$ as given in Eq. (4b)) and the effect of L_2 on η_s through the tapering of the waveguide at the corner. As a result, the curves of η_s , first, increase to the maximum values and then decrease back. For example, in the single curved bending ($L_1 = 0$, Fig. 3(a)) at the bend angles of 3° , 5° and 7° , the simulated values of η_s maximize at 0.99, 0.89 and 0.64, respectively.

Additionally, as it is shown in Fig. 3(b)–(d) an appropriate increase in L_1 causes an increase in the normalized transmission coefficient in the case of double curved bending. Comparison of the curves in Fig. 3(b), (c) and (d) that correspond to the cases of $L_1 = 80$, 120 and 160 μm , respectively, determines the role of L_1 in η_s . This is due to decrease of $\Delta\Phi$ by increase of L_1 (based on Eq. (4b)) and consequently raising the normalized transmission coefficient as explained in Section 2. For example, in the case of $L_1 = 80 \mu\text{m}$, we observed maximum values of η_s to occur at 0.996, 0.99 and 0.83 corresponding to the angles of

Table 1

Comparison of the lengths of the structures with almost equal normalized transmission coefficient

Bend angle θ	Design parameters of double curved bending, μm (proposed method)	Design parameters of linear tapered bending, μm	Double curved bending (proposed method)		Linear tapered bending	
			Device length (μm)	η_s	Device length (μm)	η_s
3°	$L_2 = 50$ $L_1 = 120$	$D_b = 5$ $D_c = 1$	170	0.99	343	0.994
5°	$L_2 = 140$ $L_1 = 120$	$D_b = 9$ $D_c = 3$	280	0.99	410	0.99
7°	$L_2 = 170$ $L_1 = 160$	$D_b = 12$ $D_c = 5$	330	0.96	389	0.923

Fig. 4. The wave propagation along (a) a double curved bending of 7° and (b) a conventional bending of 7° .

3°, 5° and 7°, respectively, which are greater than those of the case when $L_1 = 0$. This point implies that the mismatch due to a wide-angle bend can be compensated by a larger L_1 . From the results shown in Fig. 3 (b)–(d), we observe that the slope of η_s versus L_2 for a wider angle θ is more than that of a smaller θ . This means that the larger values of L_1 and L_2 can compensate the mismatch of the larger θ more effectively which in turn confirms the discussion given in Section 2.

So far, we have found out that increases in the lengths of L_1 and L_2 improve the normalized transmission coefficient (lower loss) of the bend. However, the penalty of this improvement is a longer device. This is also confirmed by comparing the values of η_s of the structures which are shown in Fig. 3. In order to have a better understanding of the roles of L_1 and L_2 on the length of the wide-angle waveguide bend, few more devices are analyzed. The “linear tapered bending” structure is used as the reference. We designed and simulated the double curved bends at angles 3°, 5° and 7° with various L_1 and L_2 lengths for having almost the same normalized transmission coefficient η_s values as those of the linear tapered bends counterparts. The results are presented in Table 1, where D_b and D_c which are defined in Section 1 and shown in Fig. 1(b) are used to indicate the amount of taper in the linear tapered bending. The results in Table 1 indicate that the proposed method is more effective than the other methods in reducing the device dimension at small bending angles. Table 1 compares the lengths and η_s of three proposed structures with those of the linear tapered counterparts, where a significant reduction in the lengths of the double curved devices especially at lower angles of bending is occurred. In fact, the method of this paper either improves the transmission coefficient or shortens the device length.

In order to show the effect of a double curved bending on guiding the optical power through the bend, a device of this type at the wide-angle bend of 7° is simulated by BPM and the field distribution in the whole path is illustrated in Fig. 4(a). Also, this illustration is compared with that of a conventional bend in Fig. 4(b). We see the high quality of wave guiding in the optimized double curve bend by its significant low loss and radiation.

4. Conclusion

We proposed two simple structures of low-loss wide-angle tapered circular bends: single curved bending and double curved bending. We observed significant improvements in the normalized transmission coefficient and the

device dimension of the proposed structures. The normalized transmission coefficient of the proposed structures obtained as large as 0.996, 0.992 and 0.983 at the bend angles of 3°, 5° and 7°, respectively. By applying the proposed method we reduce the structure lengths in the scale of 42%, 32% and 16% with respect to the “linear tapered bending” at the corresponding bend angles of 3°, 5° and 7°, respectively. In summary, the implementation of the proposed approach results in significant improvement in both the transmission power and the dimension of the optical planar waveguide bends.

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